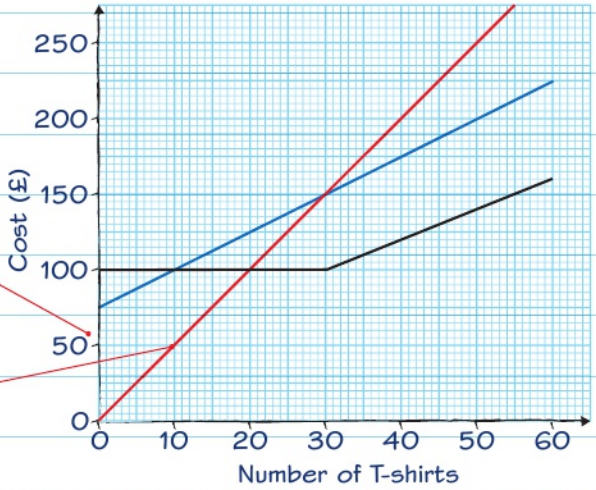
Real-life graphs

You can draw graphs to explain real-life situations. This graph shows the cost of buying some printed T-shirts from three different companies.

Be careful when reading scales on graphs. 10 small squares represent £50 so each small square represents £5.

Terry's T-shirts would be cheapest if you were ordering 10 T-shirts.



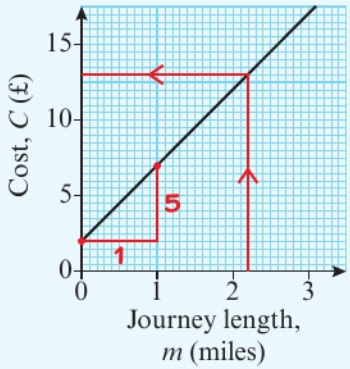
TERRY'S T-SHIRTS
No minimum order
£5 per shirt

SHIRT-O-GRAPH
Set-up cost £75
£2.50 per shirt

PAM'S PRINTING
Special offer 30 shirts for £100
Additional shirts just £2

Worked example

This graph shows the cost, C (£), of a taxi journey of length m miles.



Everything in red is part of the answer.

(a) Use the graph to estimate the cost of a journey of 2.2 miles. **(1 mark)**

£13

(b) The taxi company charges a fixed amount plus a cost per mile. Write down:
(i) the fixed amount
(ii) the cost per mile. **(2 marks)**

(i) £2 (ii) £5

The additional cost per mile is the **gradient** of the graph.

Interpreting the gradient

You might need to **interpret the gradient** of a straight-line graph:

- Straight-line graphs have a **constant gradient**.
- The gradient tells you how much the **vertical** variable increases when you increase the **horizontal** variable by one unit.
- Pay close attention to the scale on the axes – don't just count squares on the graphs.

There's more about gradients on page 37.

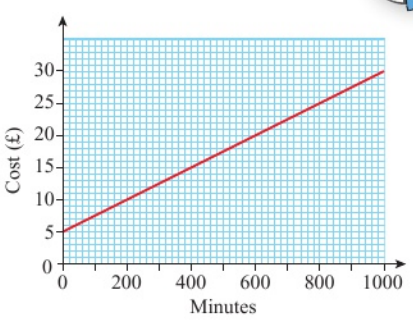
Now try this

A mobile phone company charges a fixed monthly price plus a cost per minute.

(a) What is the fixed monthly charge? **(1 mark)**

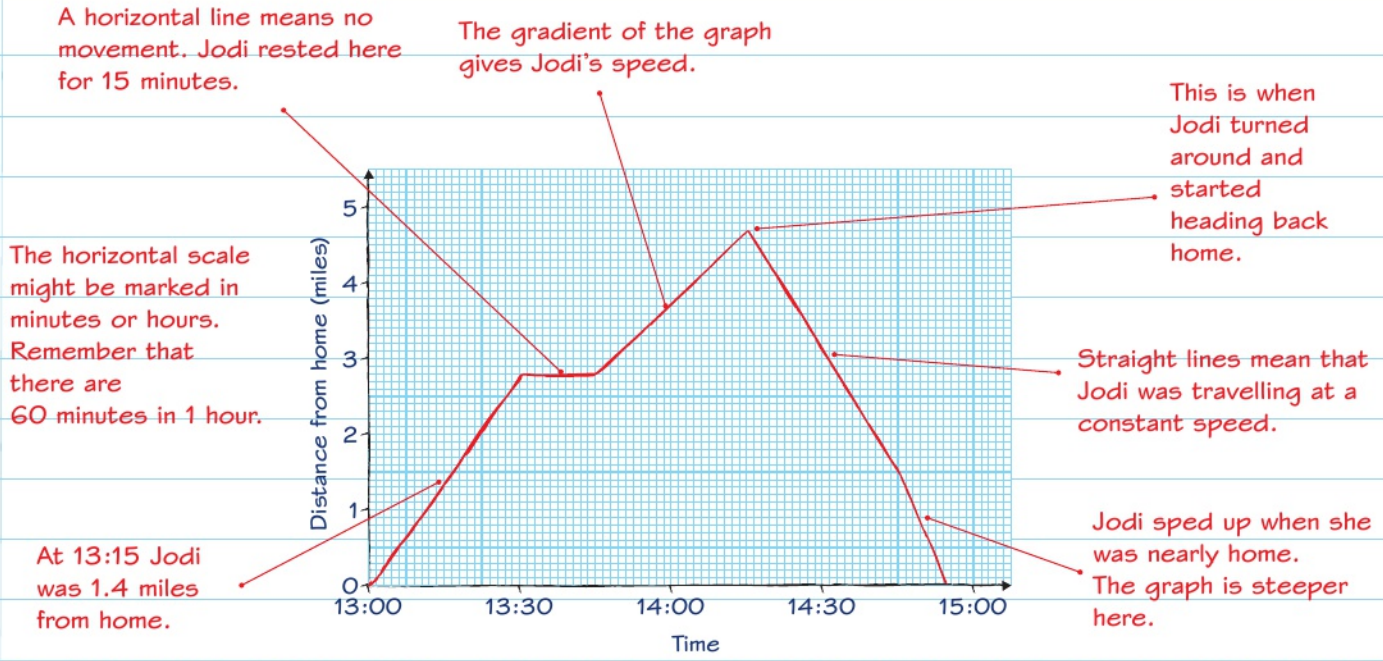
(b) What is the total cost if you use 400 minutes in one month? **(1 mark)**

(c) The phone company claims that 'For every 10 minutes you only pay an extra 25p.' Is the phone company's claim true? Give a reason for your answer. **(3 marks)**



Distance–time graphs

A **distance–time** graph shows how distance changes with time. This distance–time graph shows Jodi's run. The shape of the graph gives you information about the journey.



Worked example

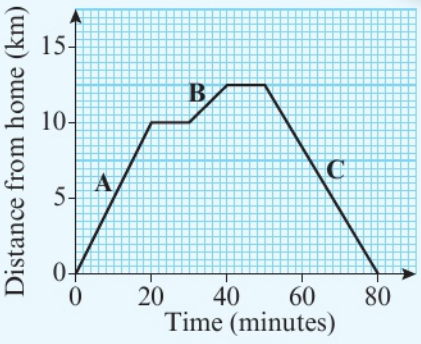
This distance–time graph shows Nick's bike ride.

(a) For how long did Nick rest in total? **(1 mark)**

20 minutes

(b) During which section of his journey was Nick travelling fastest? Give a reason for your answer. **(2 marks)**

A. Because that is where the graph is steepest.



The flat sections of the graph are when Nick is resting. Read the scale on the graph carefully.

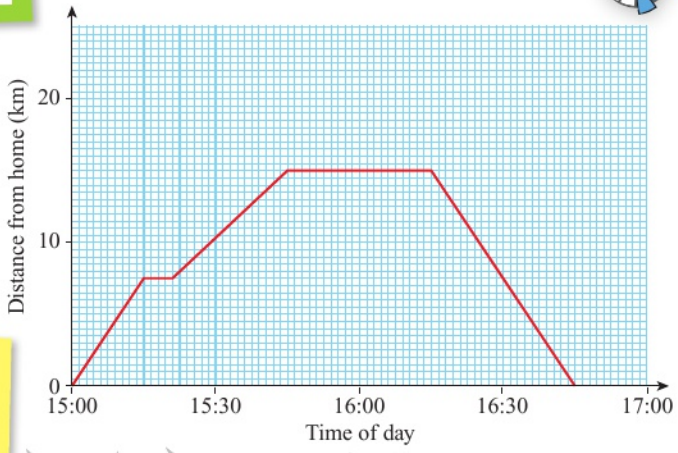
Now try this

Worked solution video

Christina rode her bike to a friend's house. She rested once on the way, had coffee at her friend's house, then rode home.

(a) How long did Christina spend at her friend's house? **(1 mark)**

(b) Work out Christina's average speed for her return journey. **(2 marks)**

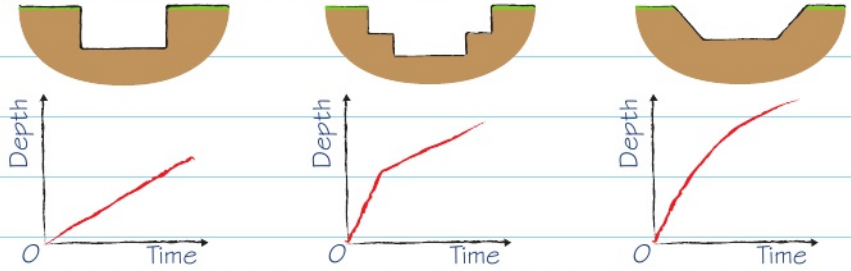


You can find the gradient of the graph, or you can use $\text{Average speed} = \frac{\text{Distance}}{\text{Time}}$

There is more about speed on page 64.

Rates of change

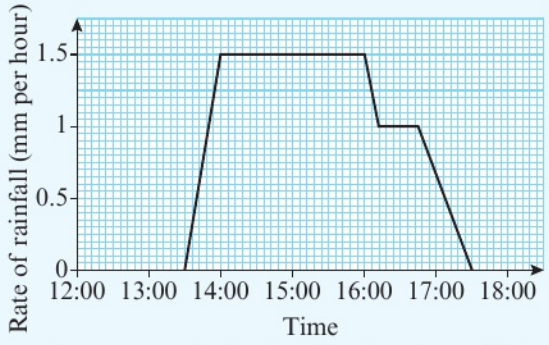
A distance–time graph shows the **rate of change** of distance with time. You can revise distance–time graphs on page 41. You can use graphs to show other rates of change. These garden ponds are being filled with water. The graphs show the rate of change of the **depth** of the water in each pond.



The narrower the section of pond, the quicker the water depth will increase.

Worked example

Supraj recorded the rate of rainfall, in mm per hour, during one afternoon.



(a) At what time did it start to rain?

13:30

(b) Supraj said, 'It stopped raining at 14:00.'
Is Supraj correct? Give a reason for your answer.

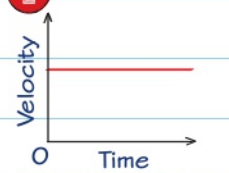
Supraj is wrong. The rain was falling steadily from 14:00 to 16:00. It didn't stop raining until 17:30.

This graph shows the **rate** of rainfall. The rain was getting harder from 13:30 to 14:00. It then fell at a steady rate until 16:00, when it started getting lighter.

Velocity–time graphs

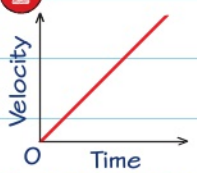
Velocity means **speed** in a certain direction. A velocity–time graph shows the rate of change of **velocity** with time. The **gradient** on a velocity–time graph tells you the **acceleration**. You need to recognise these four situations:

1



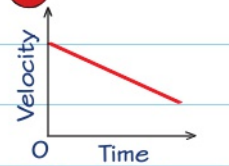
Constant speed

2



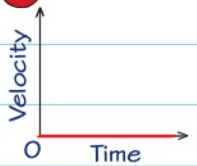
Accelerating

3



Decelerating

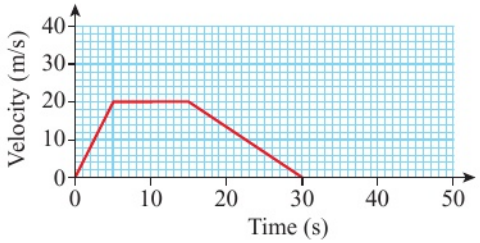
4



Not moving

Now try this

This is the velocity–time graph for a car journey.



Which of these statements are true and which are false? Give reasons for your answers.

A The car travelled at a constant speed between 0 and 5 seconds.

B The car travelled at a constant speed between 5 and 15 seconds.

C The car was accelerating between 0 and 5 seconds.

D The car was not moving between 5 and 15 seconds. **(4 marks)**